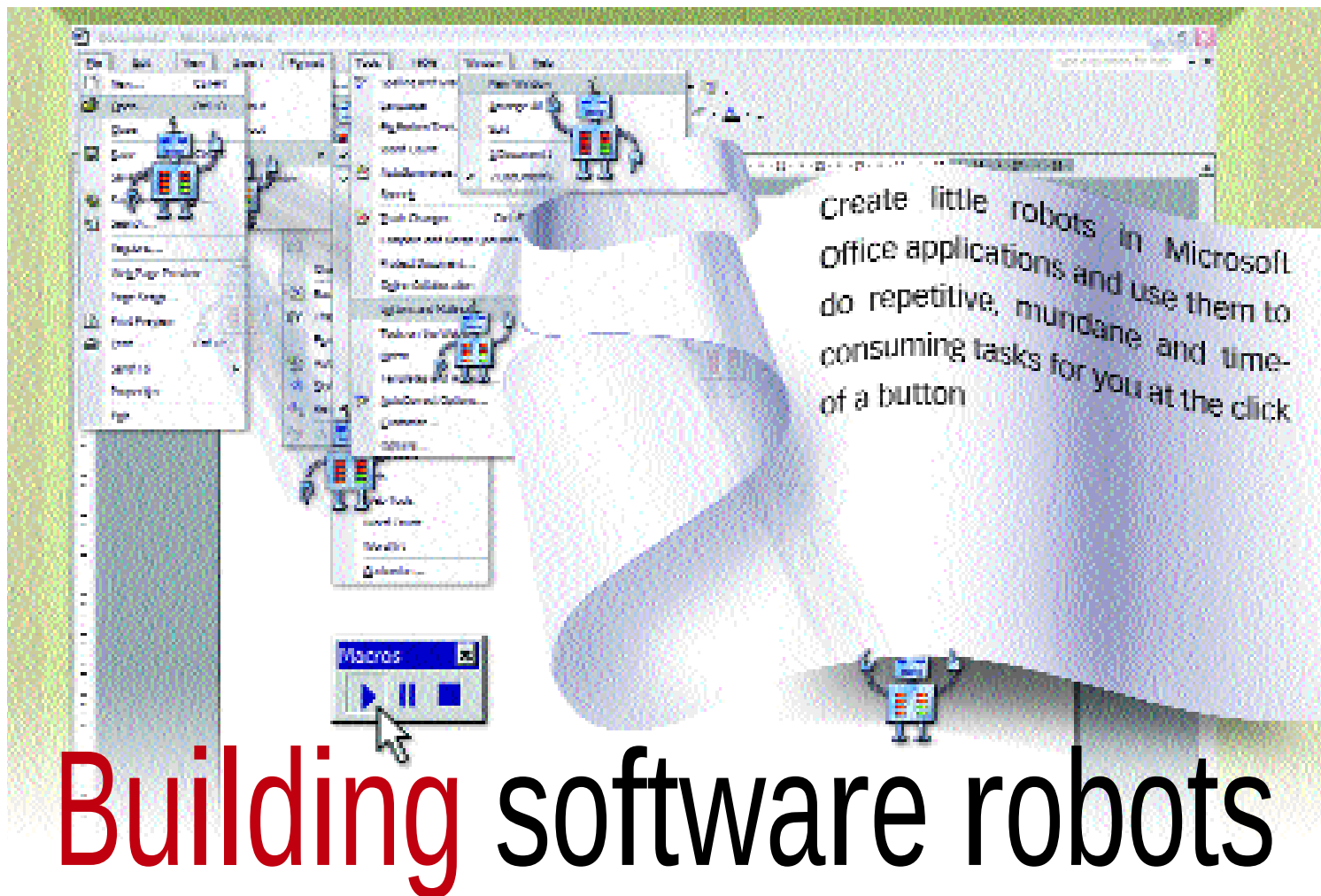




Building software robots

Kaul, the logistics manager at the head office of the courier company, walks into office and boots up his PC. It's the end of the month and he has a lot of work today—managers from all regions have sent him reports which he has to compile and pass on to the accounts department and his boss. He fires up his mail client, saves all the attachments into neatly organised folders and starts compiling all the data into one document. Having worked hard at it till a much-delayed lunch, he gives the entire document one final glance before sending it off. It wasn't a slow PC, the copy-paste operations or the two cigarette breaks that took him five hours to make the final report—he was busy changing the formatting of each document from the various regions to one consistent look and cross-checking the data in the Excel sheets.

Microsoft Office is by far, one of the most common Office suites in use today. Whether it's for writing letters, making project reports, managing balance sheets or making presentations, it is used as much at home as it is in the office. All applications in the suite highly simplify the processes of creating, updating, managing and, to some extent, even sharing content. However, often we find ourselves playing

with the same tools, doing the same things over and over again. Not only does this get thoroughly boring, it also turns out to be quite time-consuming. Microsoft has included support for macros that can automate a lot of repetitive tasks and, contrary to popular belief, the feature is extremely simple to use.

Personal robots

A macro is a series of commands and instructions grouped together, which, when run, execute these instructions automatically. They are used to perform repetitive tasks in Microsoft Office. Thus, they are actually software robots that are 'taught' a specific set of instructions once and these are mimicked in the exact same manner every time you 'switch them on'. Structurally, a macro is a block of program script, using Visual Basic for Applications (VBA). VBA is a modified version of the popular programming language, Visual Basic.

You don't need to understand a single line of code for creating basic macros. In all Office applications, you can literally 'record' a macro—the software tracks your actions (mouse clicks and keyboard entries) to generate the macro for you.

Macros are stored in the file you are working with, its template, or the default

template. Macros stored just in the file are available only while that document is open. If the macro is stored in the template for that file, it will be available to all doc-

Macro Viruses

Since macros are actually executable code, they can be malicious, just like viruses. Viruses are also executable code and can be executed independently, or via another program (an infected EXE). Similarly, macros are executed via Office documents. A simple block of code that selects all text on the document, deletes it, saves the file and closes it can be the worst nightmare for most people.

Microsoft has included a very crude virus protection technique in Office applications. Click **Tools > Options** and enable Macro Virus Protection. With this, whenever you open a document containing macros, the software will ask you whether you want to enable them or not. If it's a document that is not supposed to have macros, or if you are unsure of its source, you can disable macros and thus prevent infection. However, you should always have a good anti-virus software running that can trap and warn you of macro viruses.

